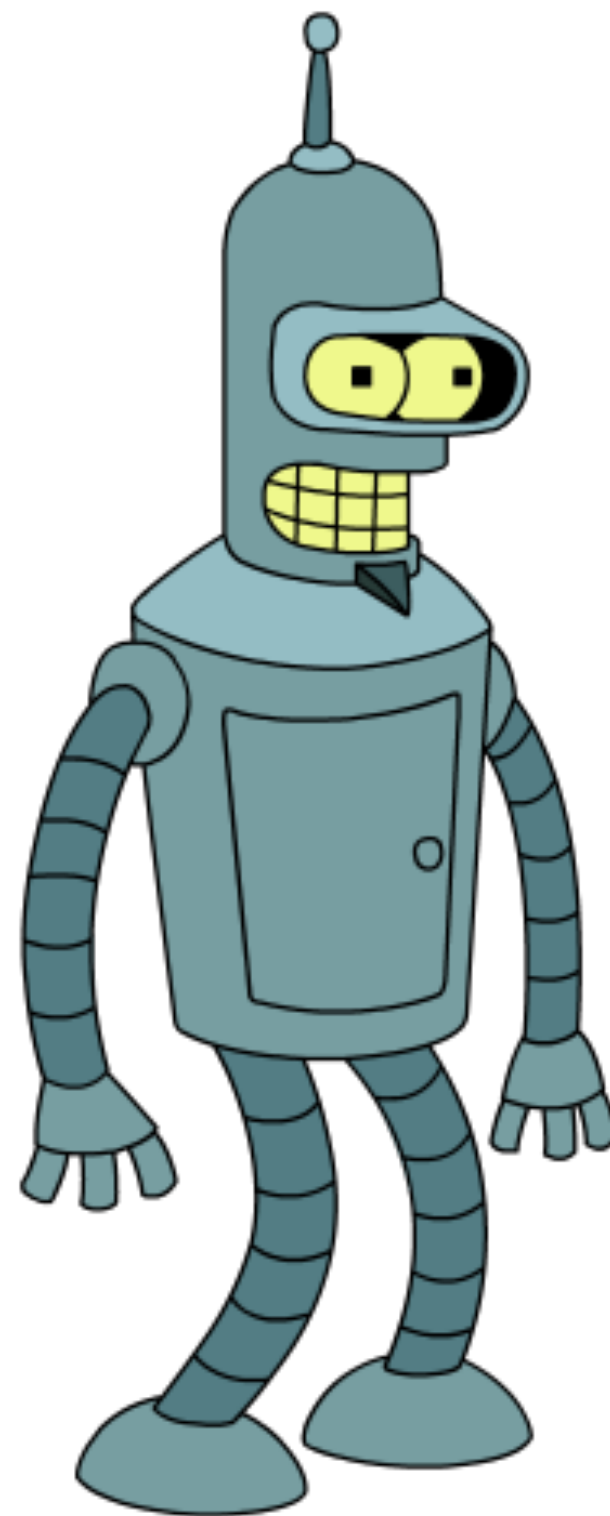


Reinforcement Learning

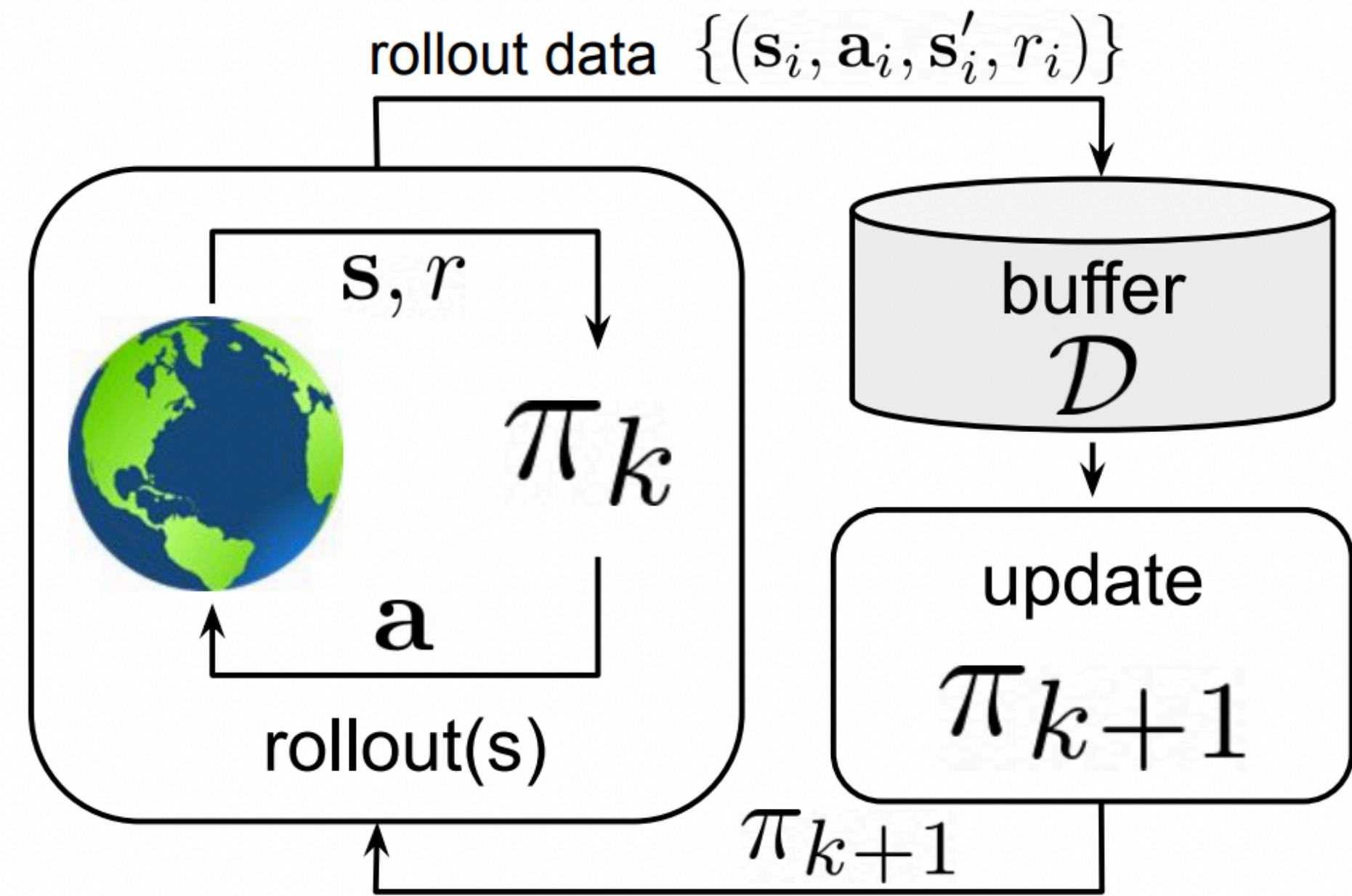
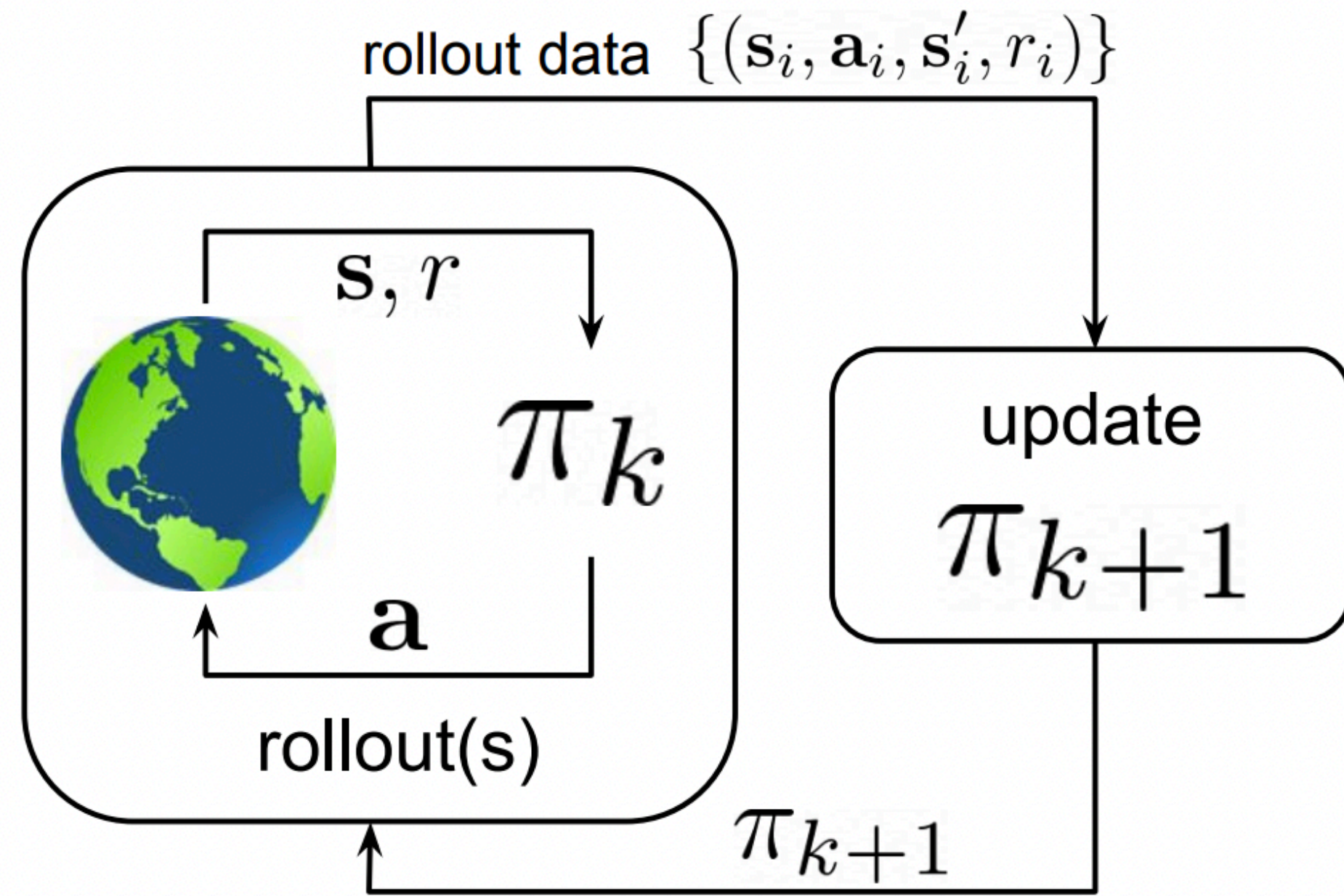
HSE, winter - spring 2026

Lecture 7: Offline RL

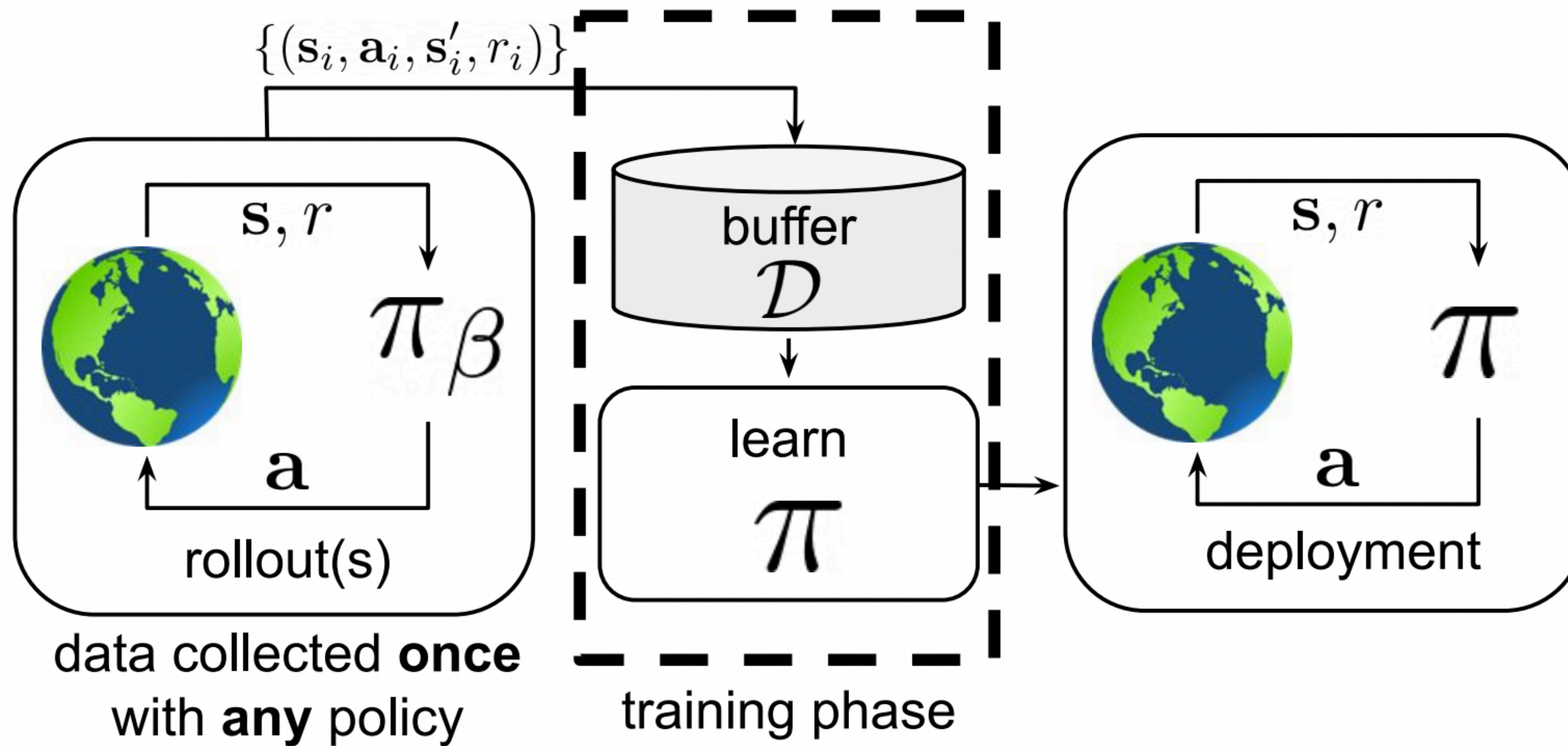


Sergei Laktionov

Recap: On-policy vs Off-policy



Offline RL



$$D = \{(s_i, a_i, r_i, s'_i)\}$$

$$1. a \sim \pi_\beta$$

$$2. s \sim d_{\pi_\beta}$$

$$3. s' \sim p(\cdot | s, a)$$

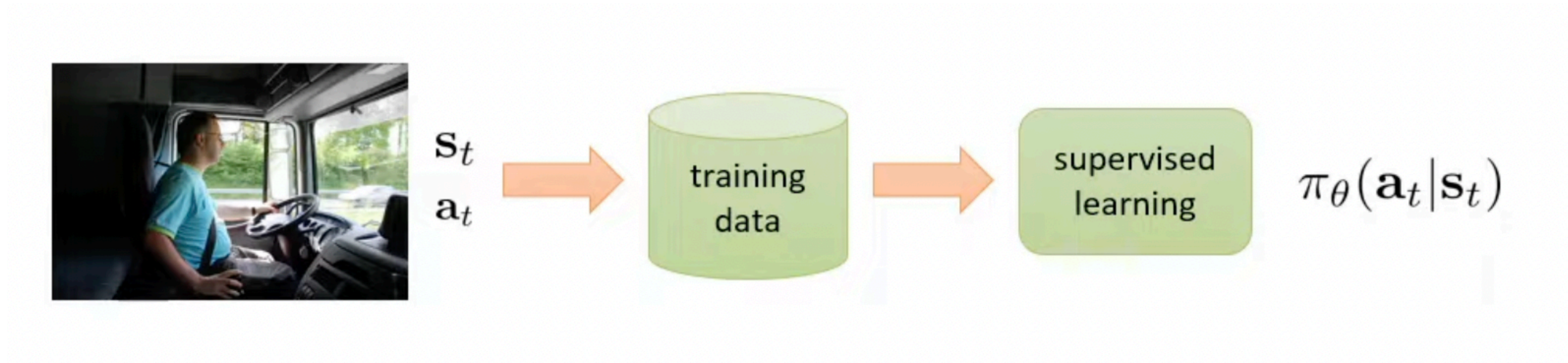
$$4. r = r(s, a)$$

Offline RL

Offline RL algorithms must do two things:

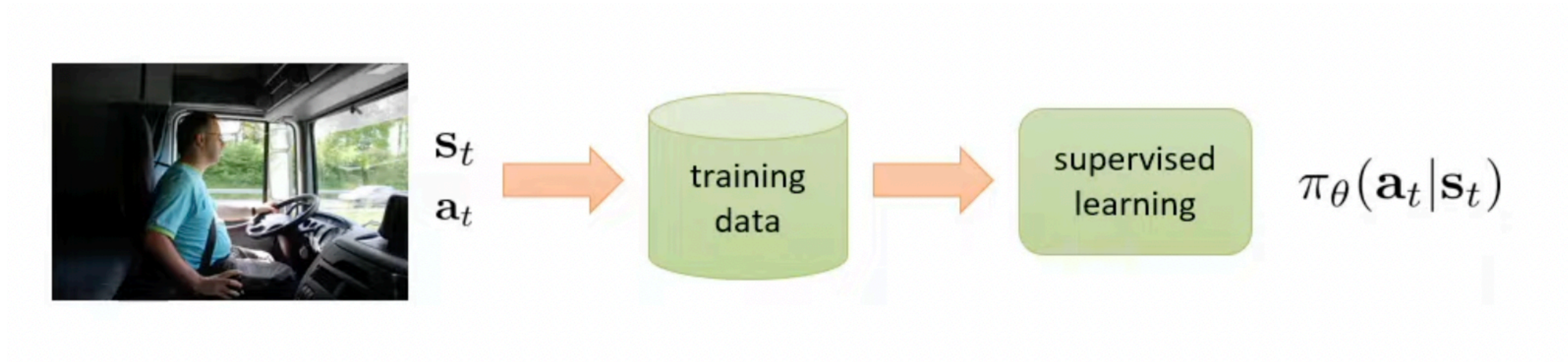
1. Maximise expected cumulative reward
2. Learn π_θ which is not very far from π_β

Imitation Learning



$$\sum_{(s,a) \in D} \log \pi_{\theta}(a | s) \rightarrow \max_{\theta}$$

Imitation Learning



$$\sum_{(s,a) \in D} \log \pi_{\theta}(a | s) \rightarrow \max_{\theta}$$

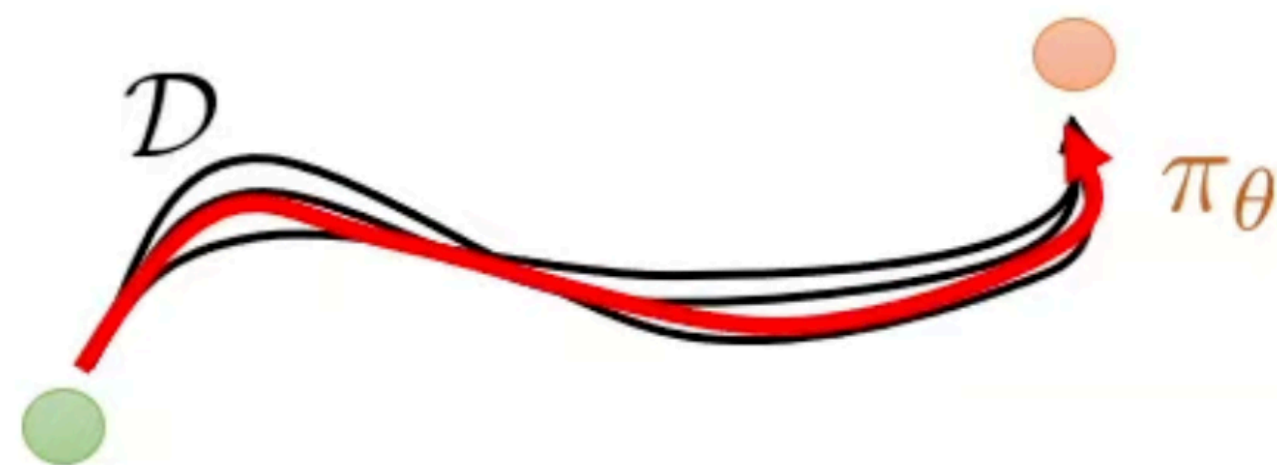
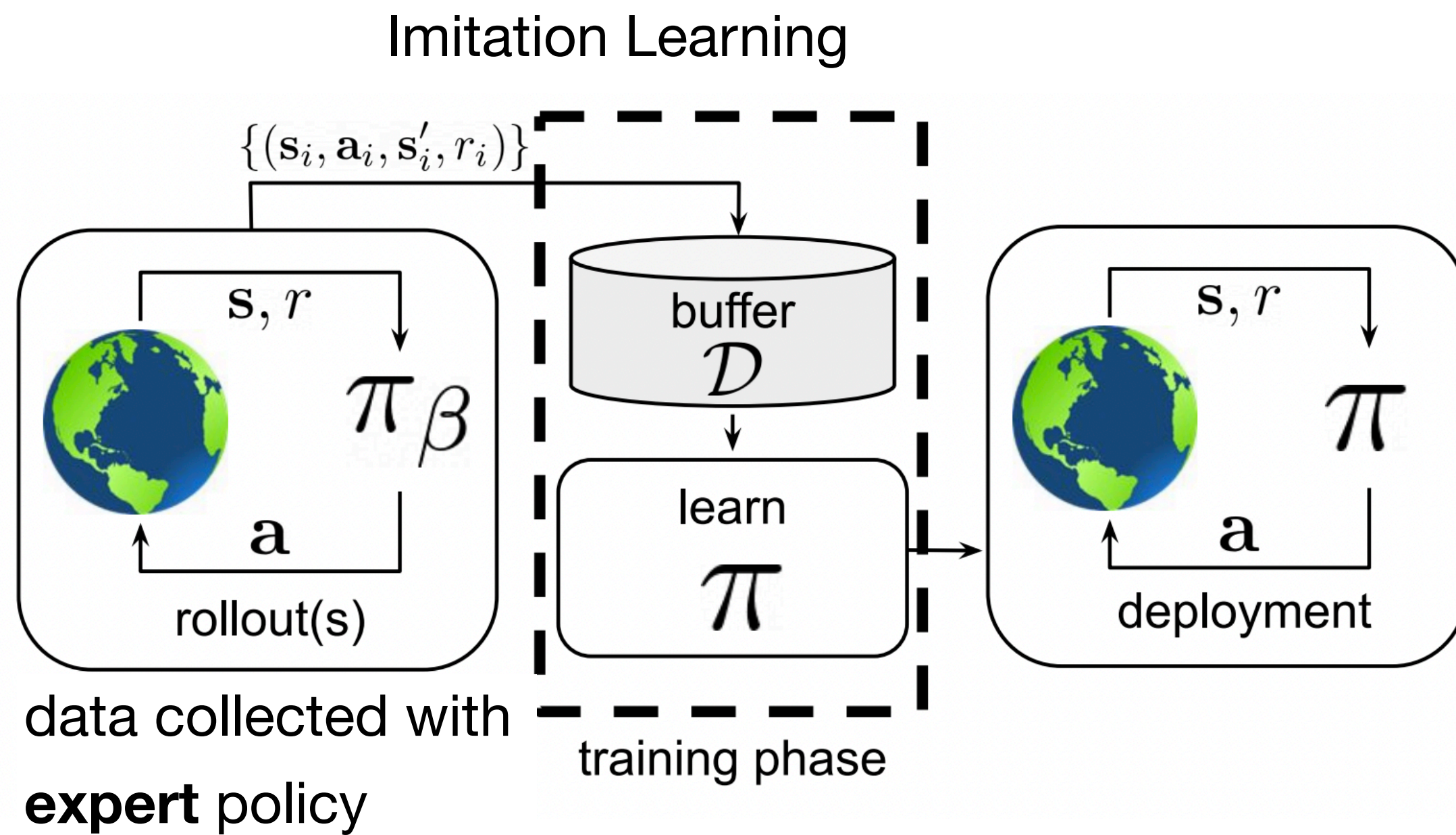
+ Very stable

+ Scalable

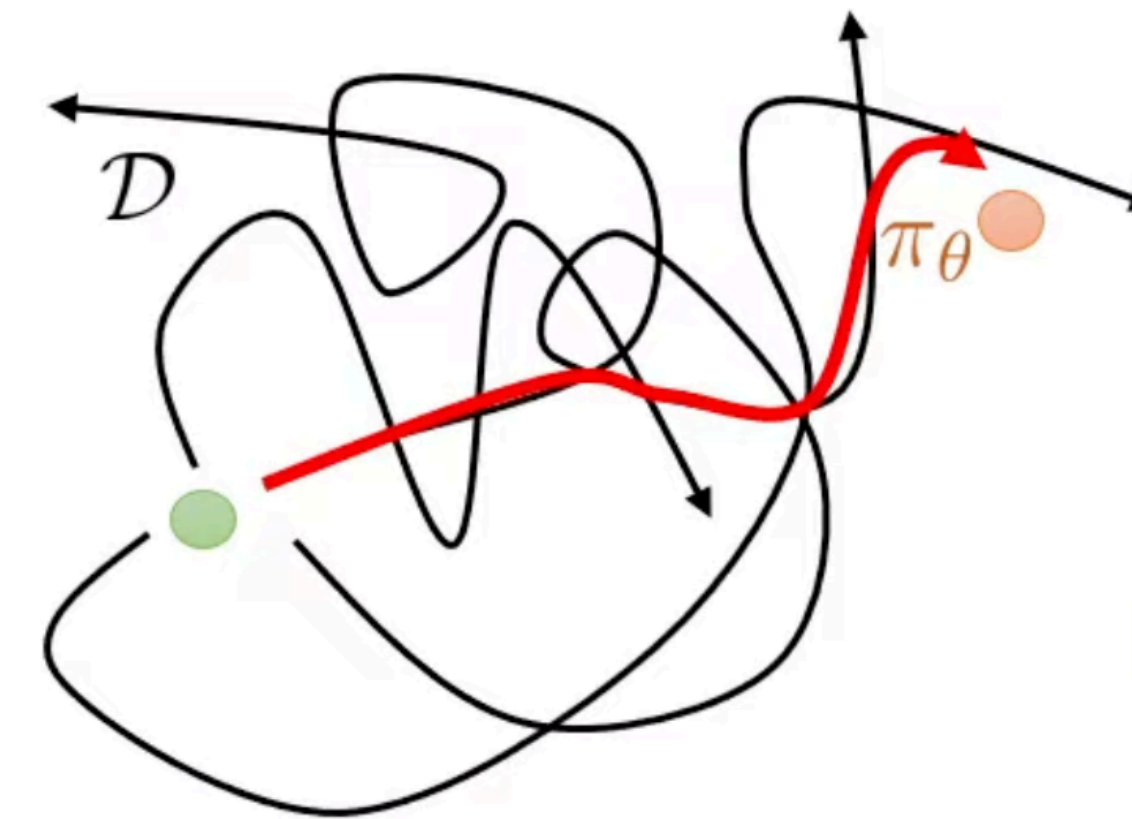
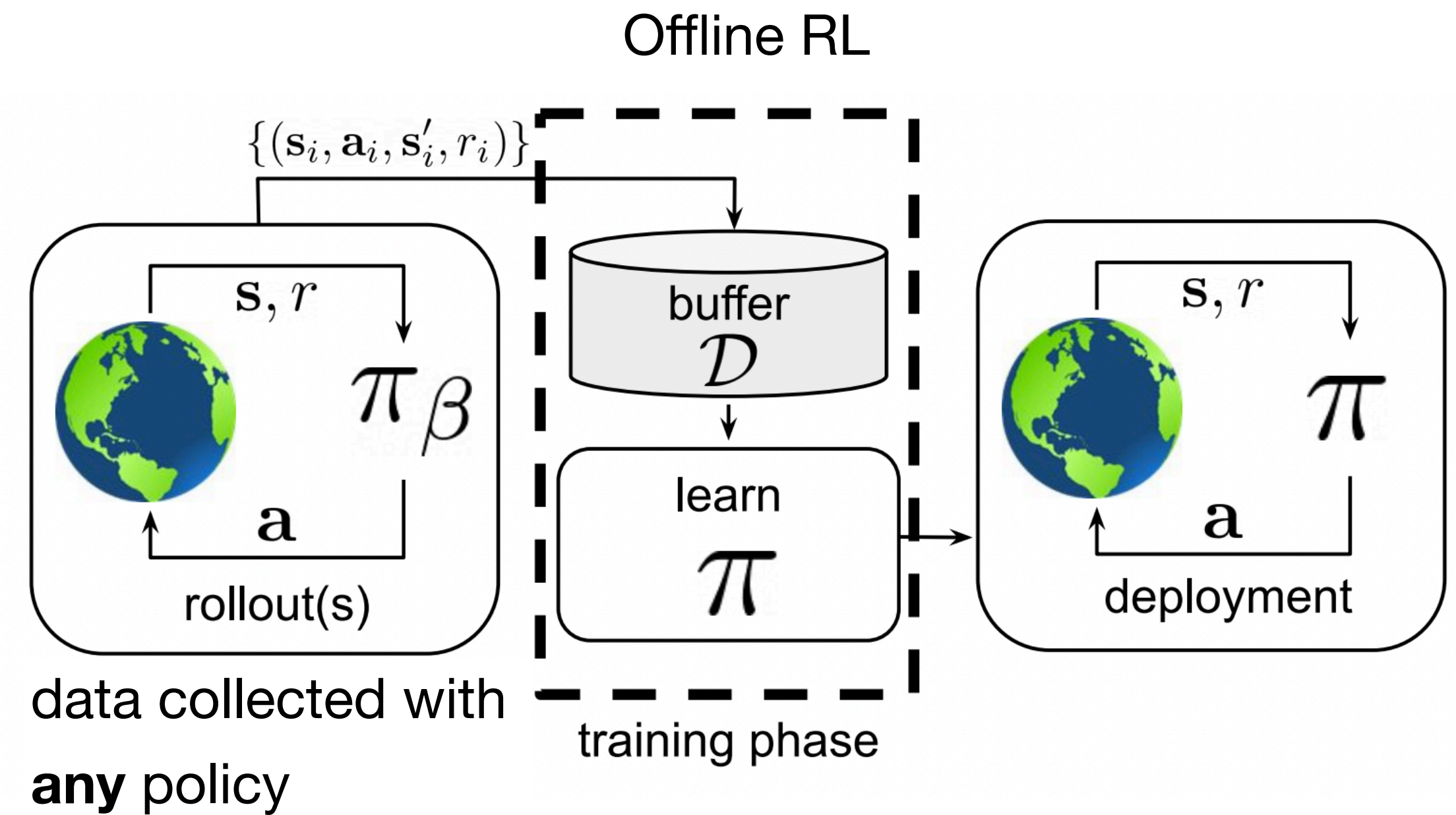
+ Easy to implement and debug

— Does not maximise the reward

Imitation Learning vs Offline RL



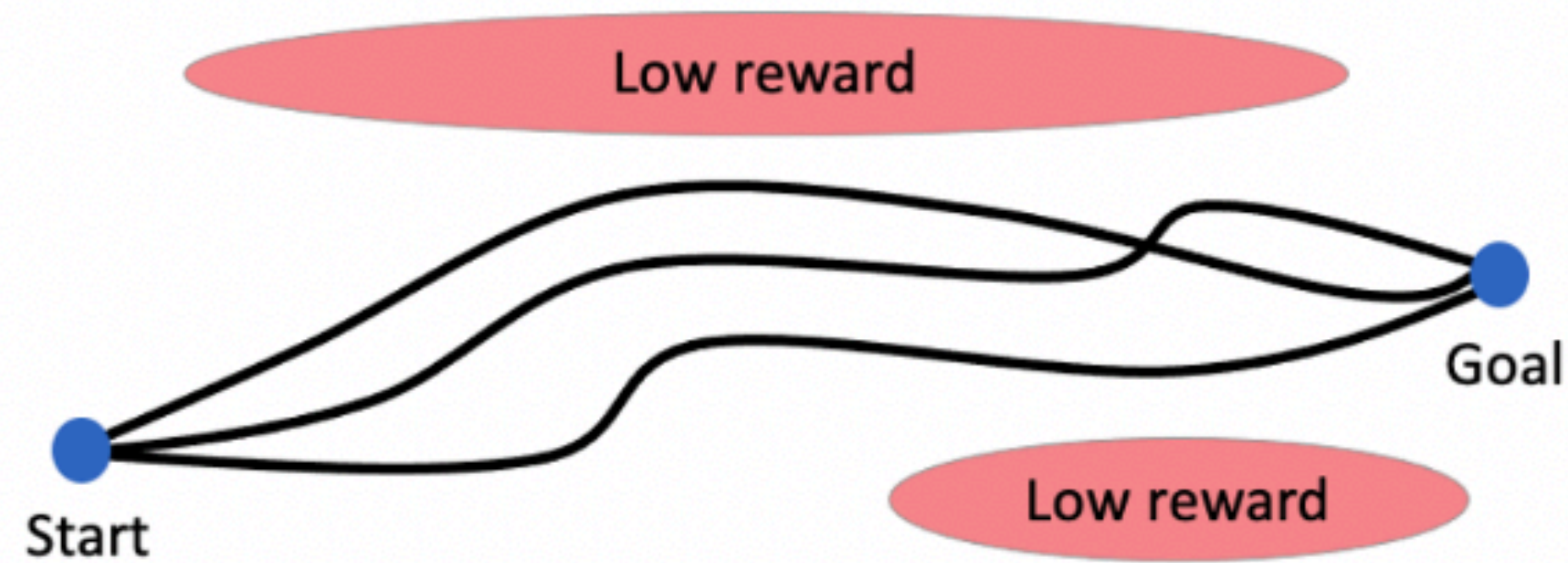
Imitate the data collected by experts



Extract the insights from noisy data

Imitation Learning vs Offline RL

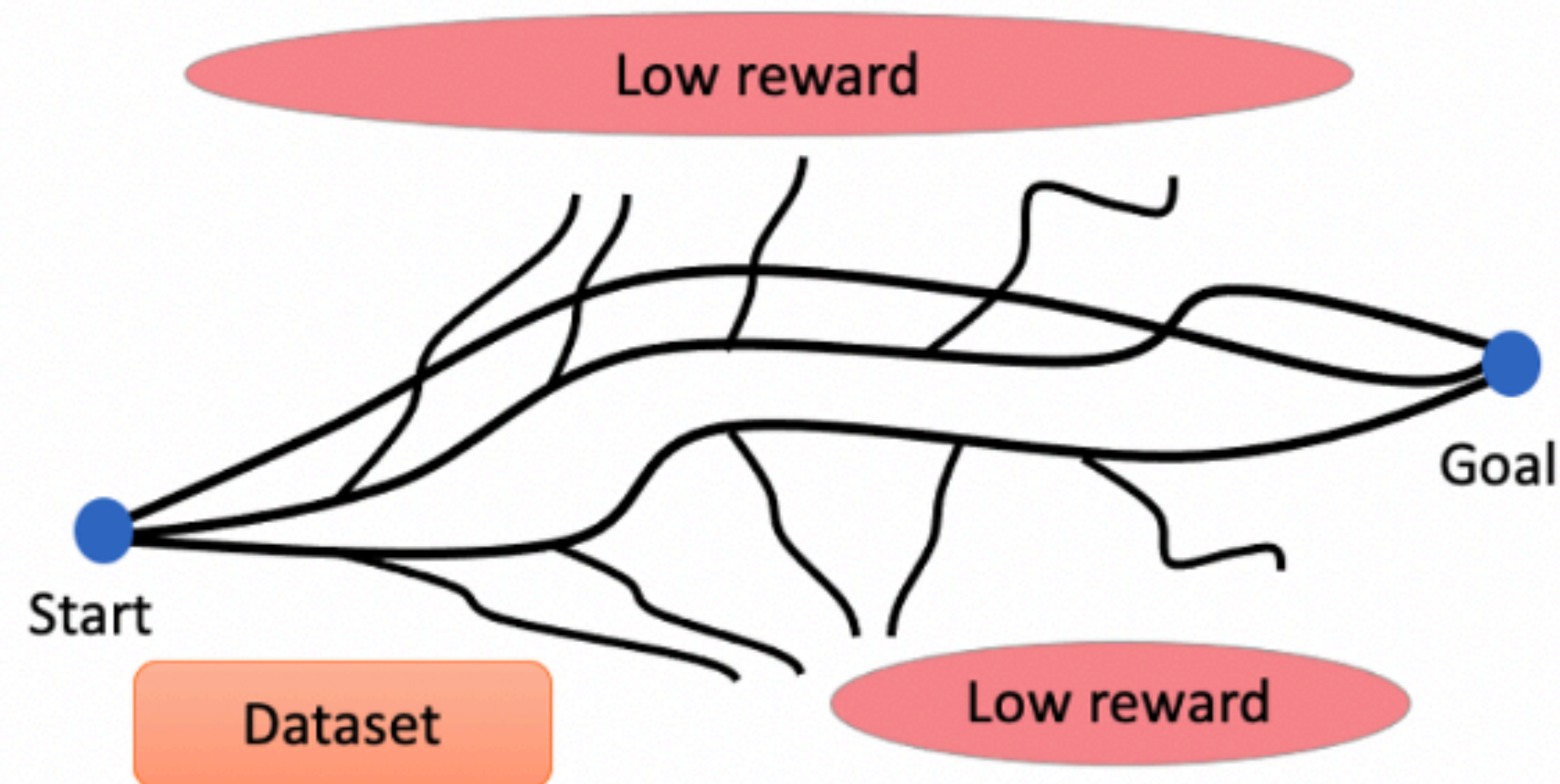
Behavior cloning on expert:



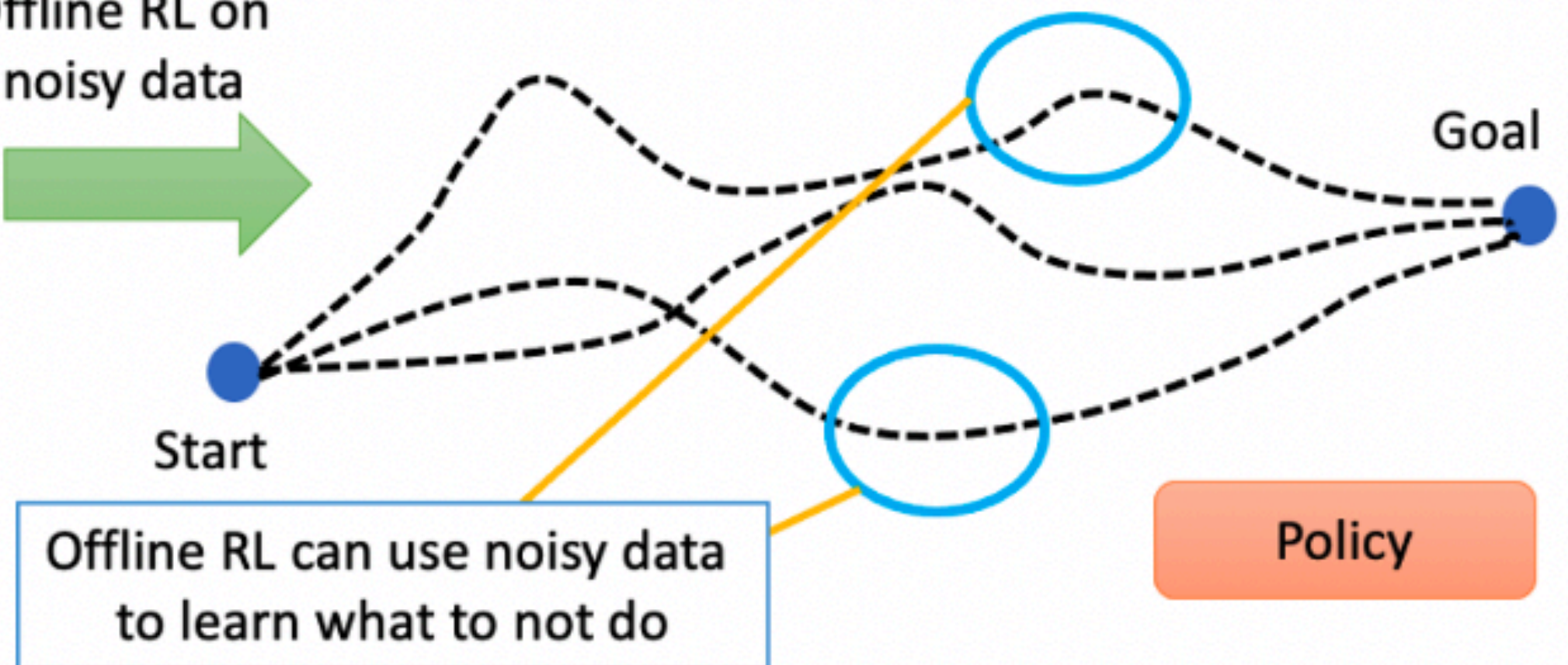
BC on
expert data



Offline RL with noisy data:



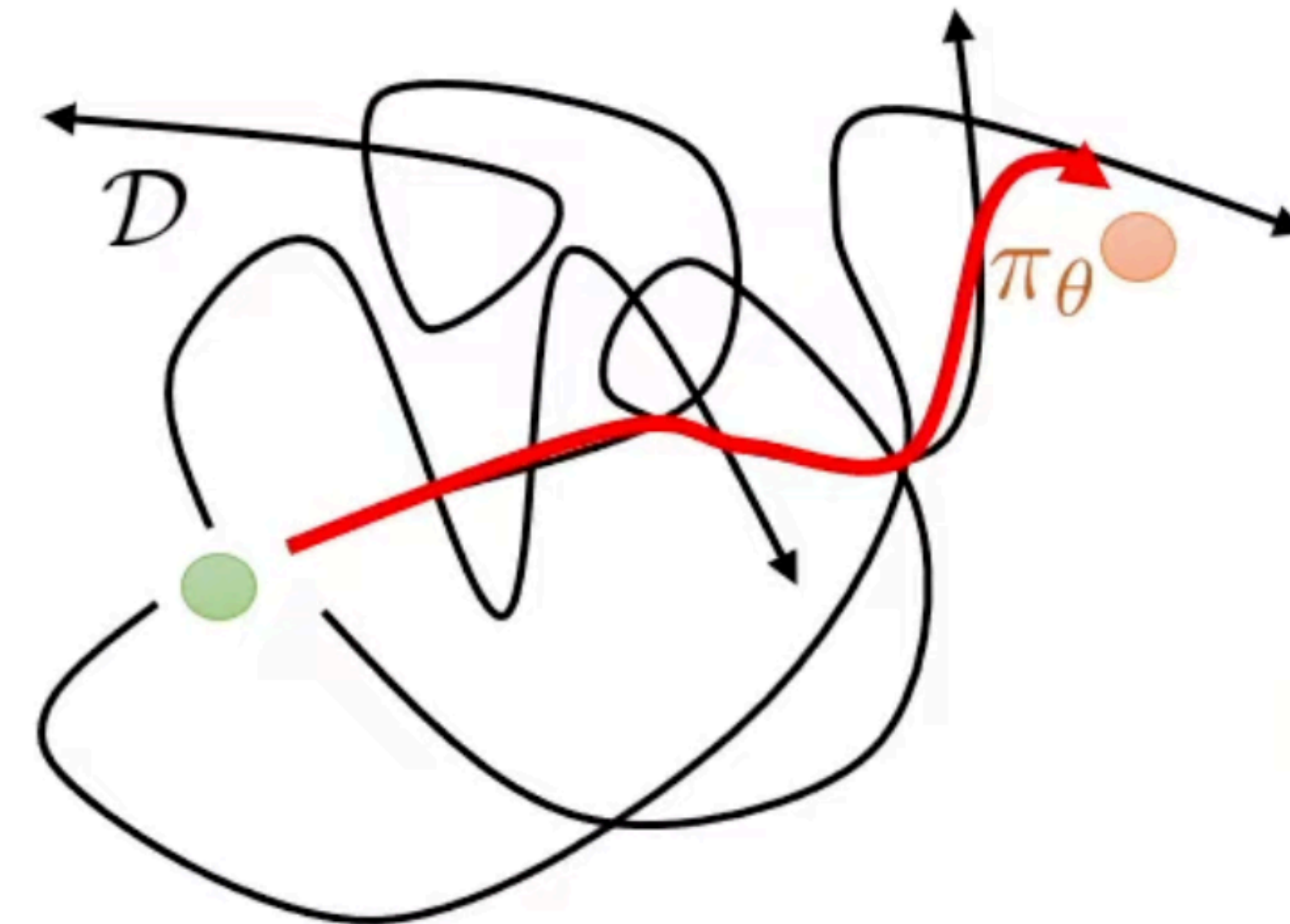
Offline RL on
noisy data



When should we prefer Offline RL over Behavioral Cloning?

Order from Chaos

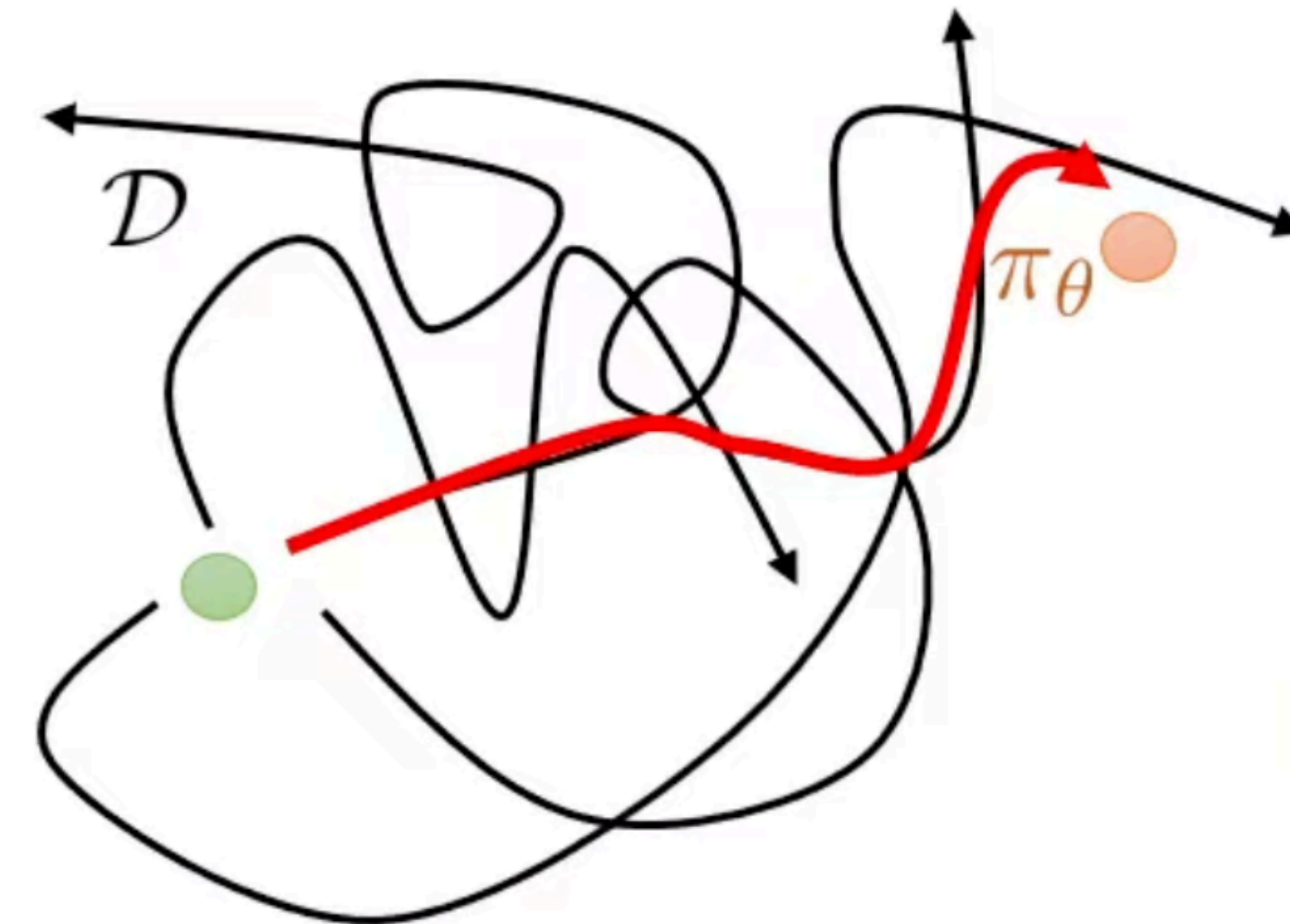
1. Extract good trajectories from both good and bad demonstrations
2. Generalise: good behaviour in one state may suggest good behaviour in another state
3. Recombine the parts of good behaviour in different scenarios



Order from Chaos

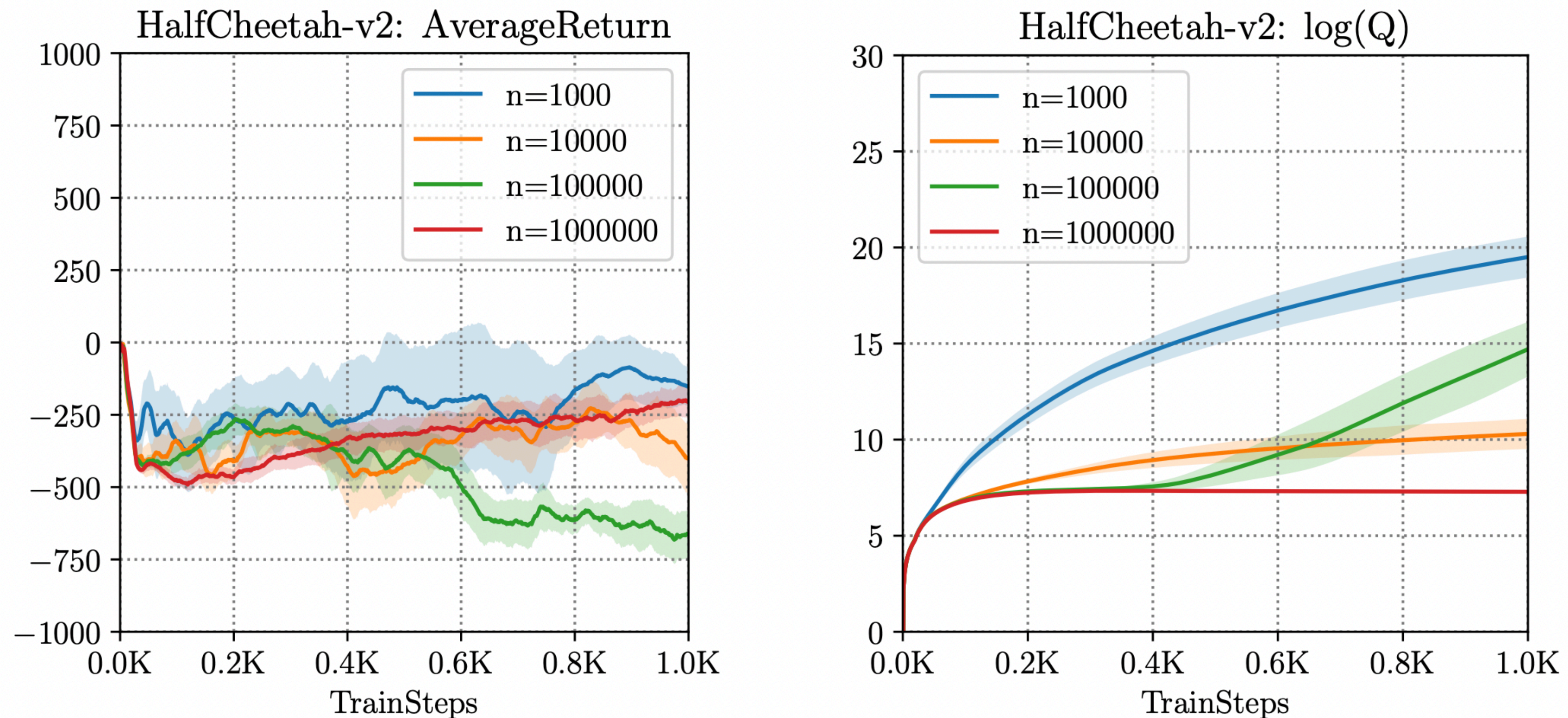
1. Extract good trajectories from both good and bad demonstrations
2. Generalise: good behaviour in one state may suggest good behaviour in another state
3. Recombine the parts of good behaviour in different scenarios

In theory, any off-policy method could be used. However they were devised under the assumption that further online interactions are possible and thus often fail.



Bootstrapping Error

Performance of SAC



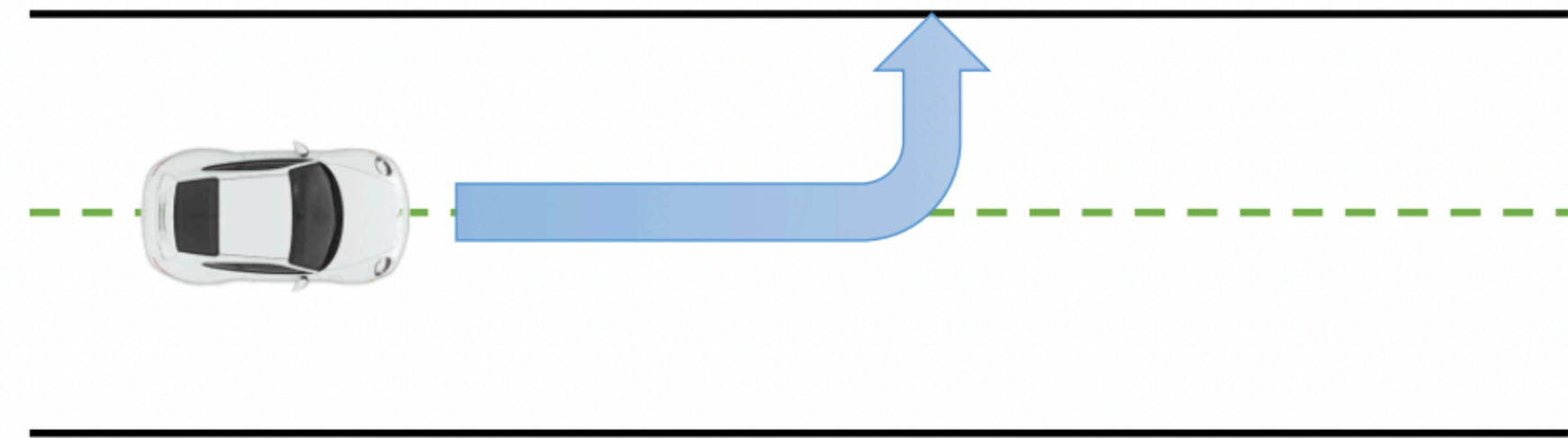
Stabilizing Off-Policy Q-Learning via Bootstrapping Error Reduction

Distribution Shift

Training data

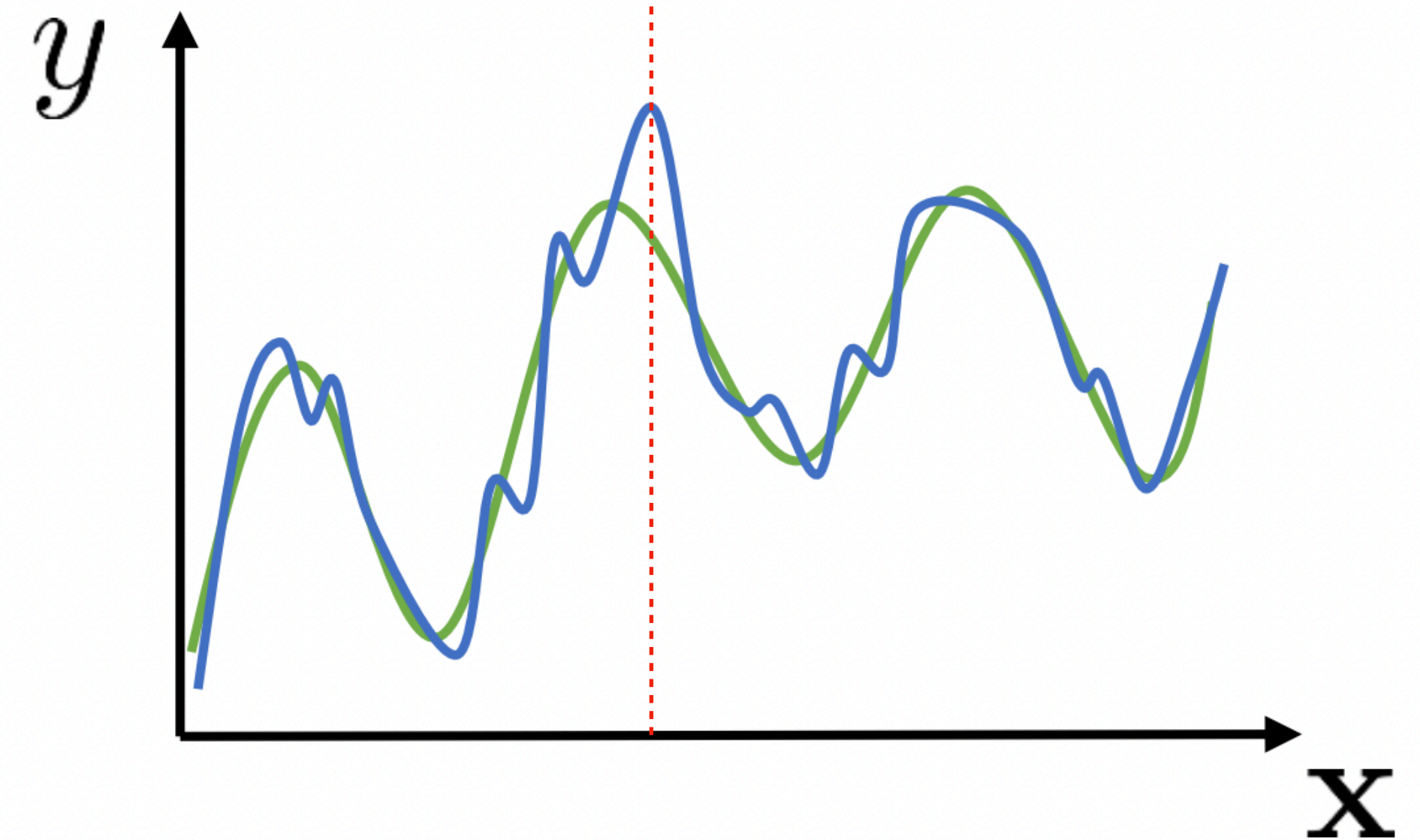


What the policy wants to do



$$y(s, a) = r + \gamma \mathbb{E}_{a' \sim \pi_{new}} [Q_{\phi}(s', a')]$$

$$\mathbb{E}_{s \sim d_{\pi_{\beta}}, a \sim \pi_{\beta}} [y(s, a) - Q_{\phi}(s, a)]^2 \rightarrow \min_{\phi}$$



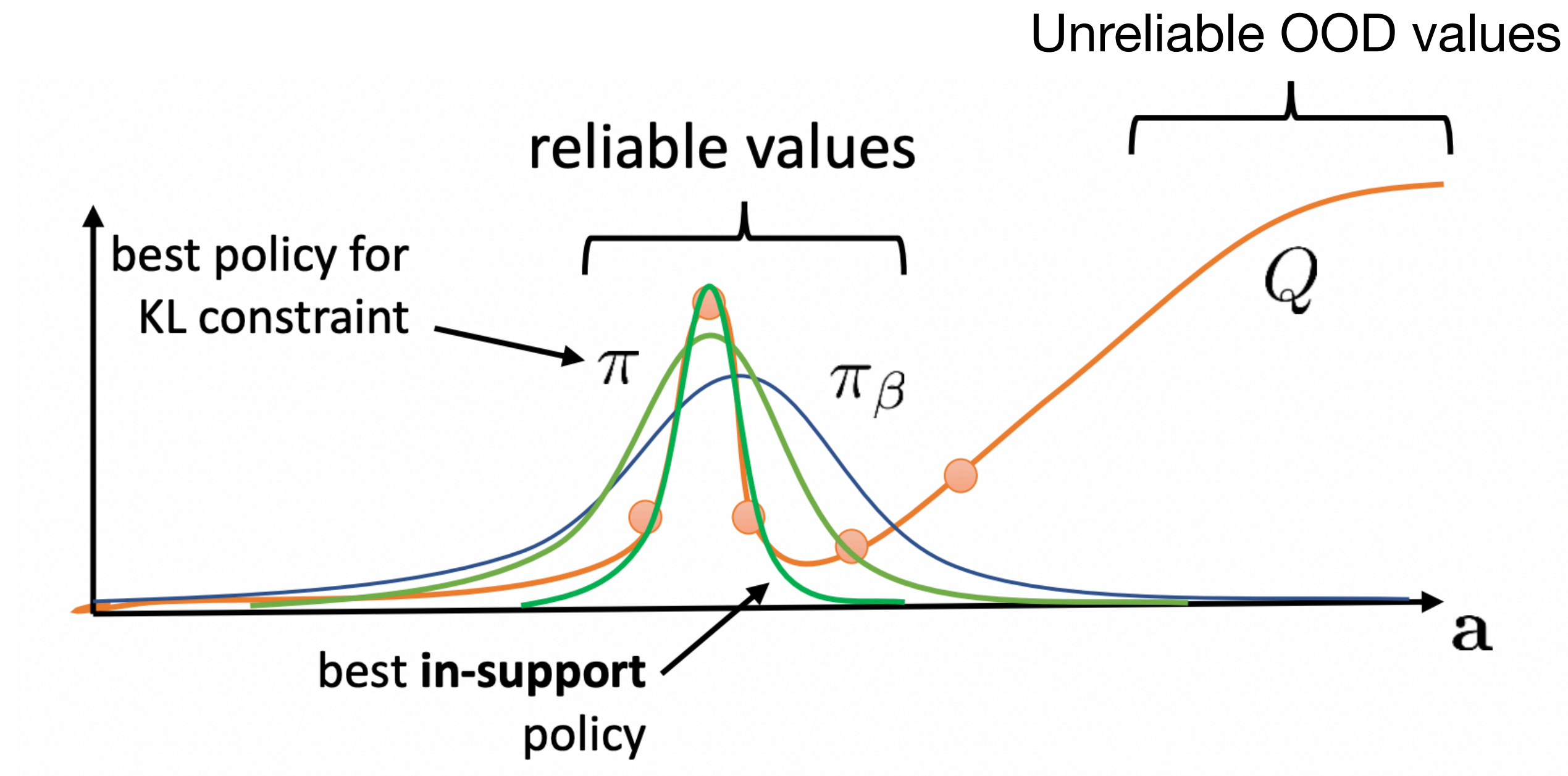
Explicit Policy Constraints

Actor objective:

$$\mathbb{E}_{s \sim D} \mathbb{E}_{a \sim \pi(\cdot | s)} [A(s, a)] \rightarrow \max_{\pi}$$

KL constraint:

$$D_{KL}(\pi(\cdot | s) || \pi_{\beta}(\cdot | s)) \leq \varepsilon$$



Explicit Policy Constraints

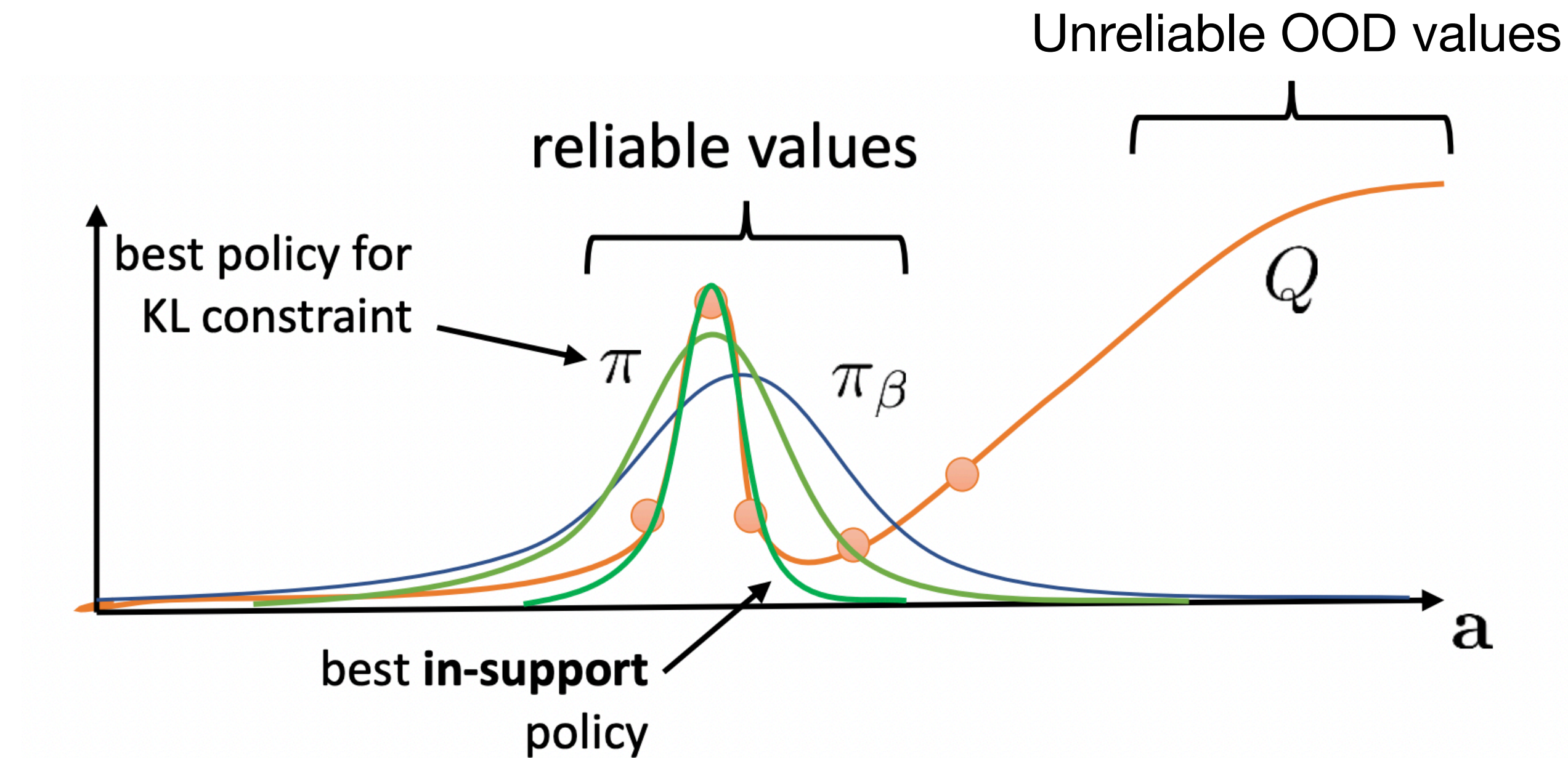
Actor objective:

$$\mathbb{E}_{s \sim D} \mathbb{E}_{a \sim \pi(\cdot | s)} [A(s, a)] \rightarrow \max_{\pi}$$

KL constraint:

$$D_{KL}(\pi(\cdot | s) || \pi_{\beta}(\cdot | s)) \leq \varepsilon$$

Support constraint: $\pi(a | s) > 0$ only if $\pi_{\beta}(a | s) > 0$



Implicit Policy Constraints

$$\mathbb{E}_{a \sim \pi(\cdot | s)}[A^\pi(s, a)] - \lambda(D_{KL}(\pi(\cdot | s) || \pi_\beta(\cdot | s)) - \epsilon) - \mu(\int \pi(a | s) da - 1) \rightarrow \max_{\pi}$$

$$\pi^*(a | s) = \frac{1}{Z(s)} \pi_\beta(a | s) \exp\left(\frac{A^\pi(s, a)}{\lambda}\right)$$

Weighted max loglikelihood:

$$\pi^{new}(a | s) = \operatorname{argmax}_{\pi} \mathbb{E}_{(s, a) \sim \pi_\beta} \left[\log \pi(a | s) \frac{1}{Z(s)} \exp\left(\frac{A^{\pi_{old}}(s, a)}{\lambda}\right) \right]$$

Advantage Weighted Actor Critic

$$\text{Actor: } \mathbb{E}_{(s,a) \sim \pi_\theta} \left[\log \pi_\theta(a | s) \frac{1}{Z(s)} \exp\left(\frac{A_\phi(s, a)}{\lambda}\right) \right] \rightarrow \max_\theta,$$

where $A_\phi(s, a) = Q_\phi(s, a) - Q_\phi(s, a'), a' \sim \pi_\theta(\cdot | s)$.

$$\text{Critic: } \mathbb{E}_{(s,a,r,s') \sim D} \left[r + \gamma \mathbb{E}_{a' \sim \pi_\theta(\cdot | s')} Q_\phi(s', a') - Q_\phi(s, a) \right]^2 \rightarrow \min_\phi,$$

Advantage Weighted Actor Critic

$$\text{Actor: } \mathbb{E}_{(s,a) \sim \pi_\theta} \left[\log \pi_\theta(a | s) \frac{1}{Z(s)} \exp\left(\frac{A_\phi(s, a)}{\lambda}\right) \right] \rightarrow \max_\theta,$$

$$\text{where } A_\phi(s, a) = Q_\phi(s, a) - Q_\phi(s, a'), a' \sim \pi_\theta(\cdot | s).$$

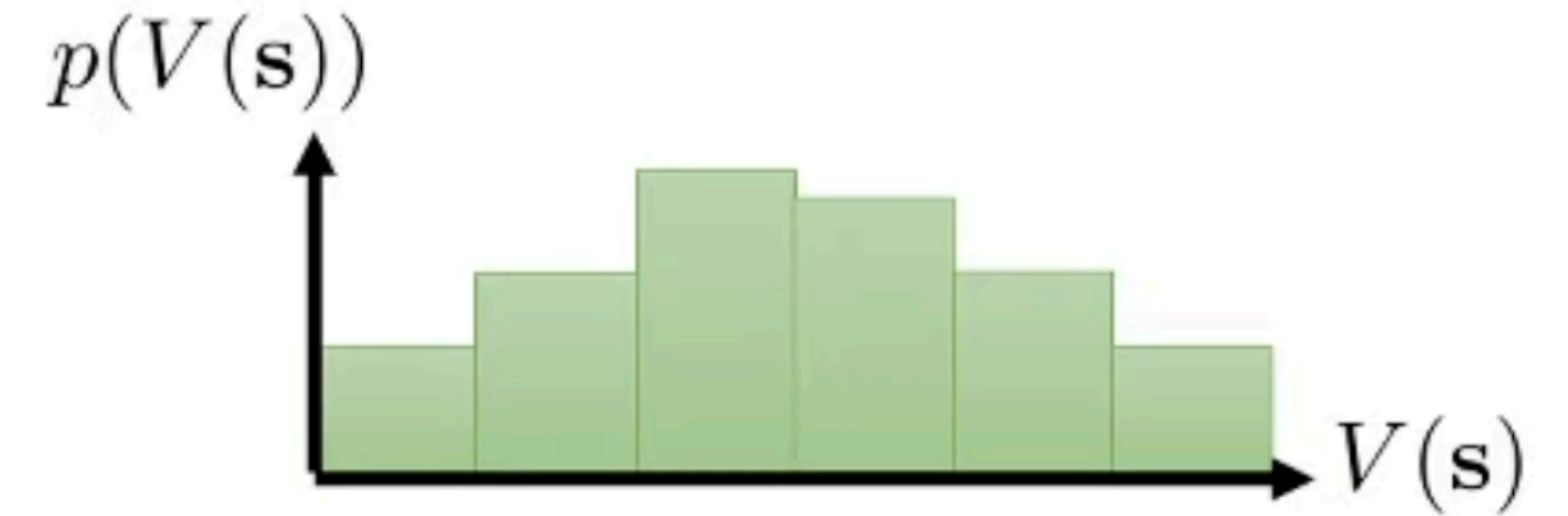
$$\text{Critic: } \mathbb{E}_{(s,a,r,s') \sim D} \left[r + \gamma \mathbb{E}_{a' \sim \pi_\theta(\cdot | s')} Q_\phi(s', a') - Q_\phi(s, a) \right]^2 \rightarrow \min_\phi,$$

Can we also avoid OOD actions?

Implicit Q-Learning

$$\mathbb{E}_{a \sim \pi(\cdot|s)} Q^\pi(s, a) = V^\pi(s)$$

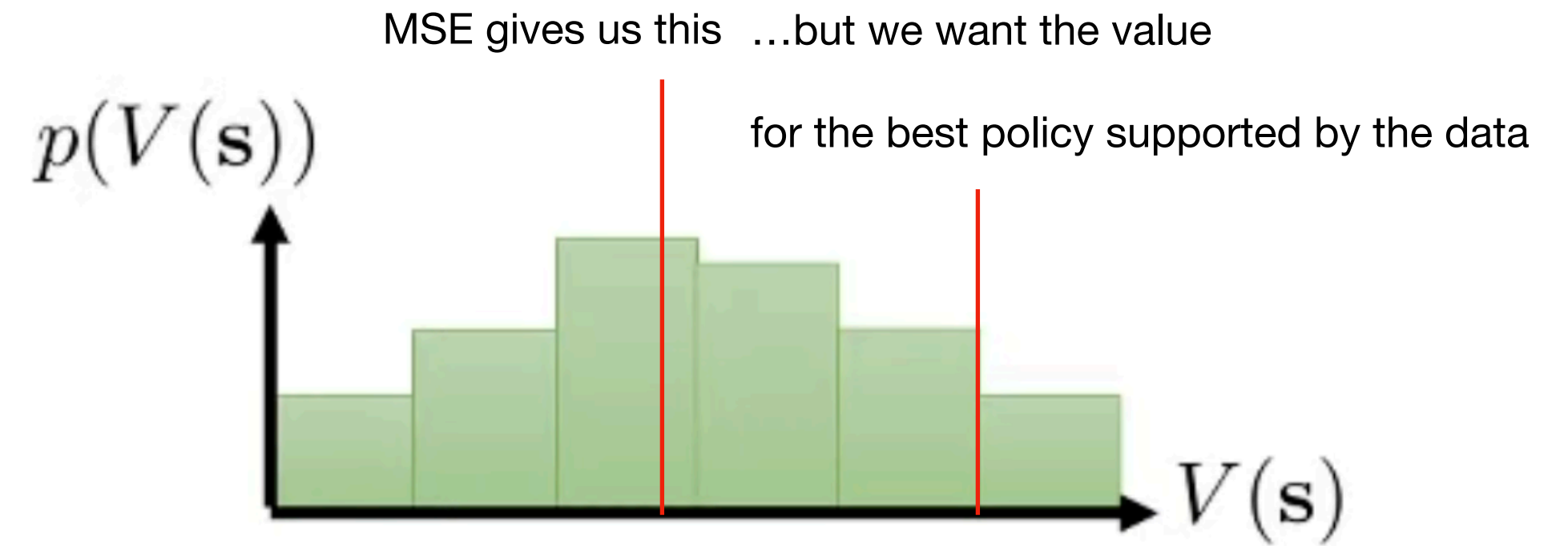
Approximate it with a neural network: $V^\pi \approx V_\psi$



Implicit Q-Learning

$$\mathbb{E}_{a \sim \pi(\cdot|s)} Q^\pi(s, a) = V^\pi(s)$$

Approximate it with a neural network: $V^\pi \approx V_\psi$



$$\mathbb{E}_{a \sim \pi_\beta} Q(s, a) \quad \max_{a: \pi_\beta(a|s) \geq \epsilon} Q(s, a)$$

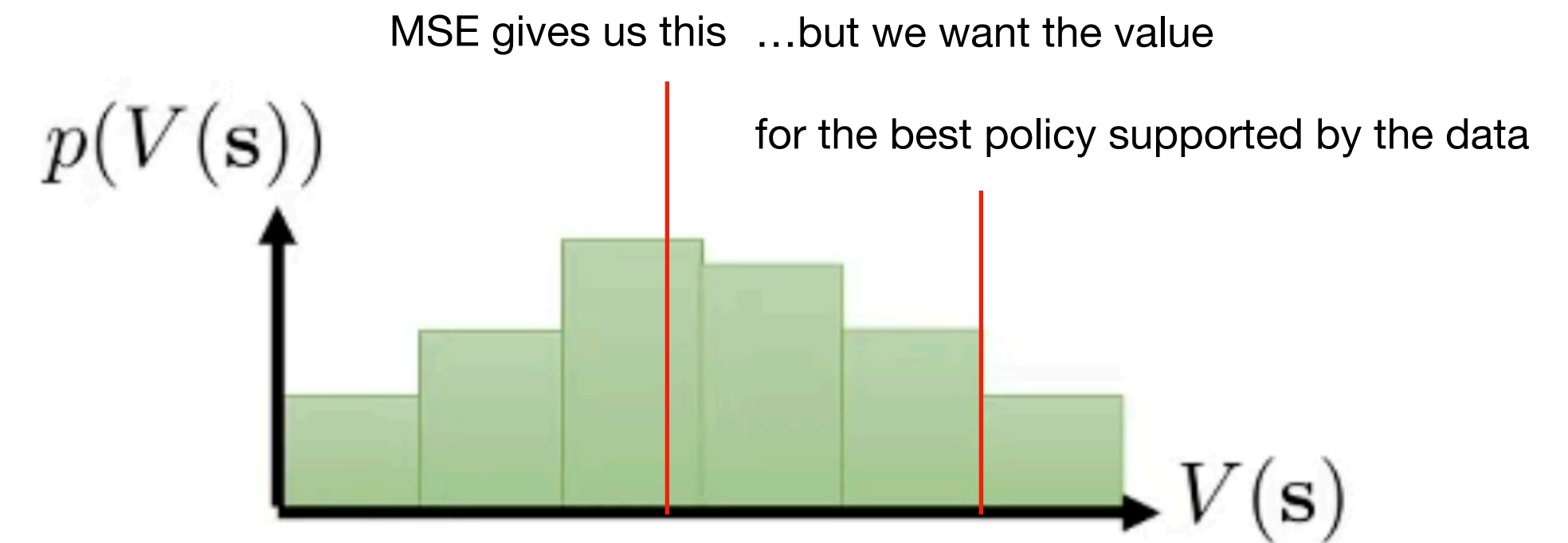
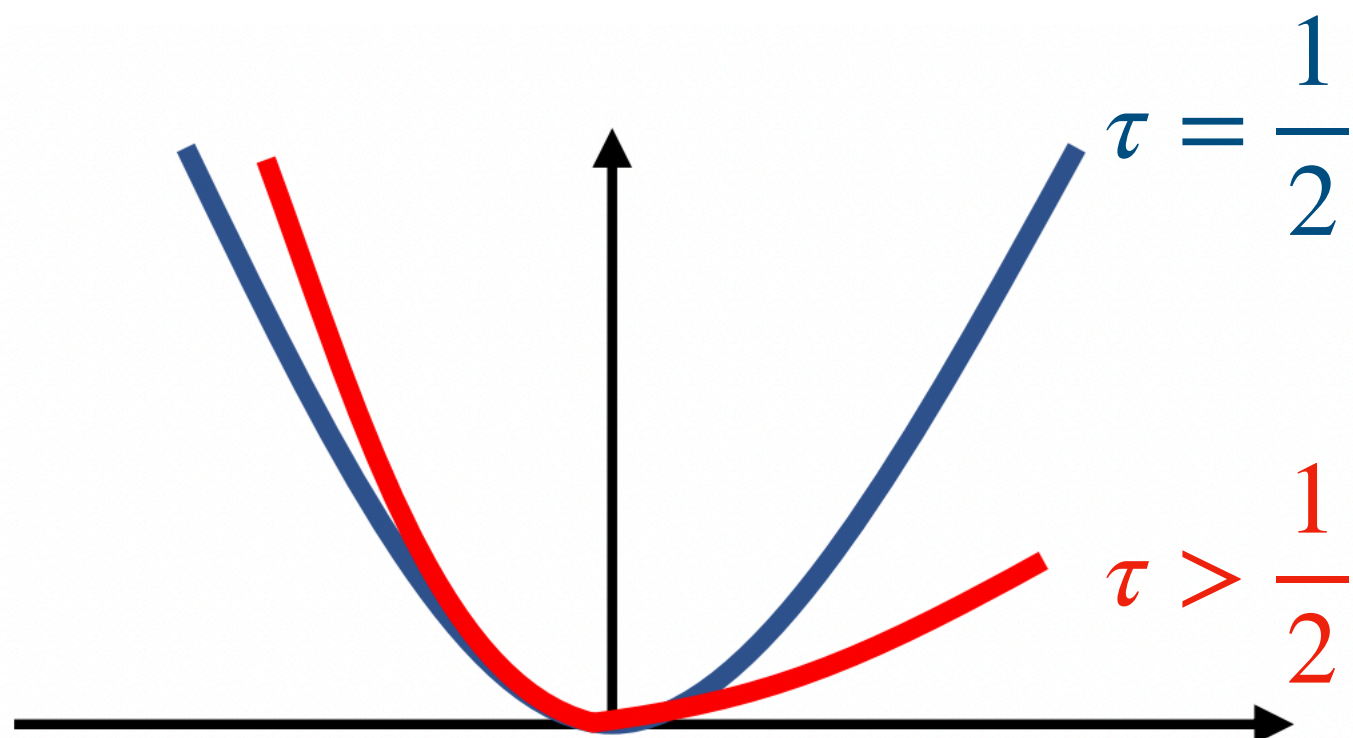
Implicit Q-Learning

$$\mathbb{E}_{a \sim \pi(\cdot|s)} Q^\pi(s, a) = V^\pi(s)$$

Approximate it with a neural network: $V^\pi \approx V_\psi$

Expectile loss:

$$L_2^\tau(u) = |\tau - \mathbb{I}(u < 0)| u^2$$

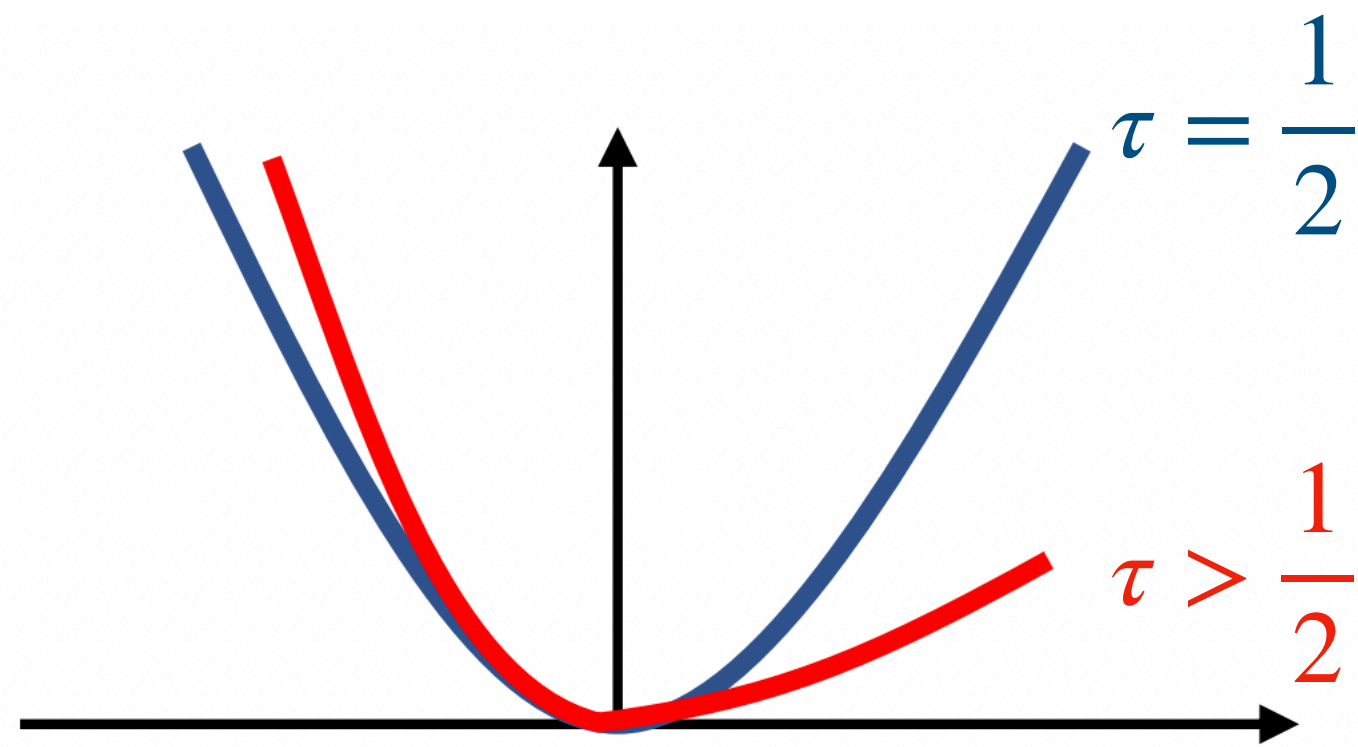


$$\mathbb{E}_{a \sim \pi_\beta} Q(s, a) \quad \max_{a: \pi_\beta(a|s) \geq \epsilon} Q(s, a)$$

Implicit Q-Learning

Expectionile loss:

$$L_2^\tau(u) = |\tau - \mathbb{I}(u < 0)| u^2$$



IQL:

$$L(\psi) = \mathbb{E}_{(s,a) \sim D}[L_2^\tau(Q_\phi(s, a) - V_\psi(s))]$$

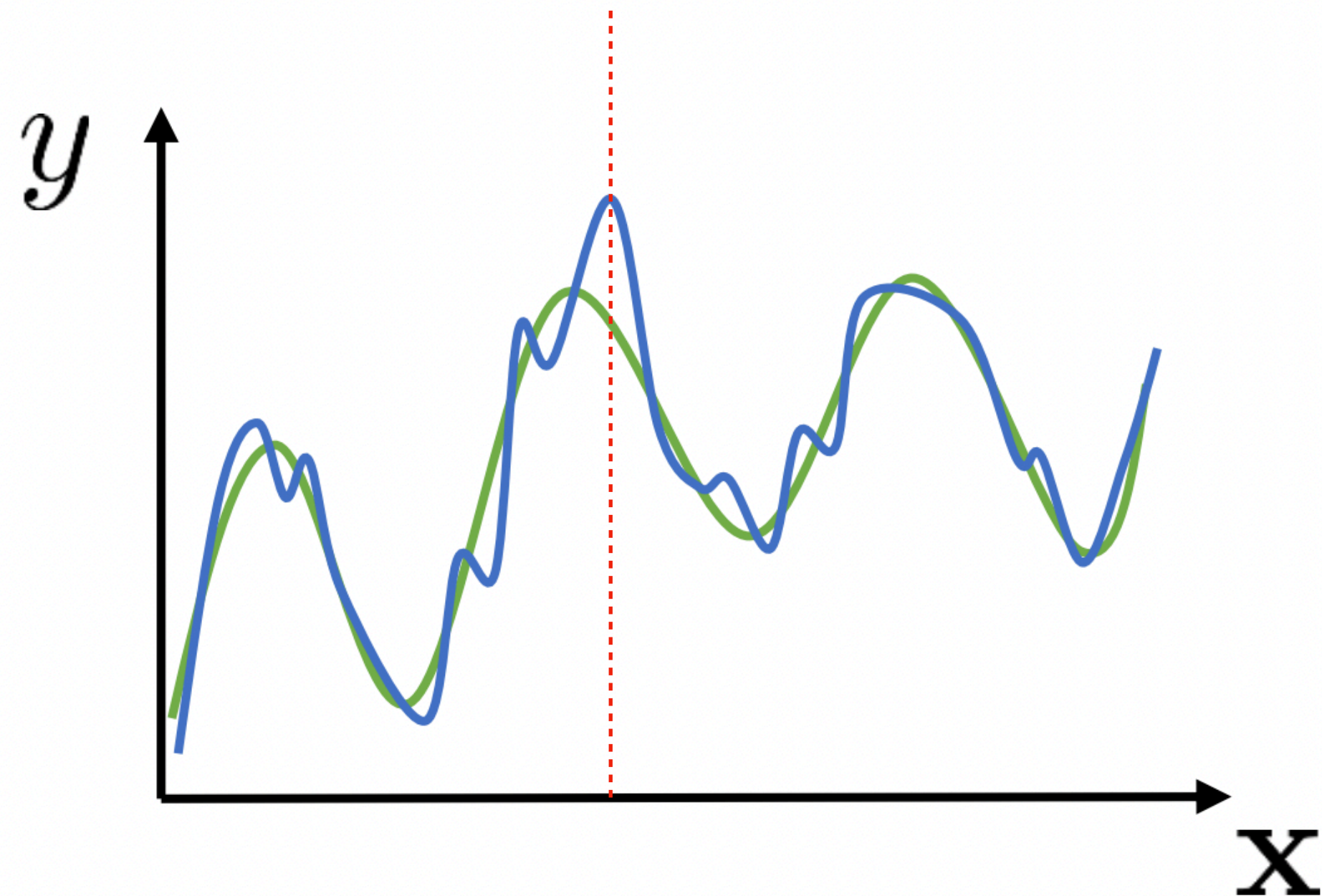
$$L(\phi) = \mathbb{E}_{(s,a,r,s') \sim D}[r + \gamma V_\psi(s') - Q_\phi(s, a)]^2$$

$$L(\theta) = -\mathbb{E}_{(s,a) \sim D}\left[\log \pi_\theta(a | s) \frac{1}{Z(s)} \exp\left(\frac{A(s, a)}{\lambda}\right)\right]$$

$$\text{where } A(s, a) = Q_\phi(s, a) - V_\psi(s)$$

Conservative Q-Learning

$$L_{CQL} = \max_{\mu} [\alpha (\mathbb{E}_{s \sim D, a \sim \mu(\cdot|s)} Q_{\phi}(s, a))] + \\ + \frac{1}{2} \mathbb{E}_{(s, a, r, s') \sim D} [r + \gamma \mathbb{E}_{a' \sim \pi} Q_{\phi}(s', a') - Q_{\phi}(s, a)]^2 \rightarrow \min_{\phi}$$



It can be shown that for large enough α

$$Q_{\phi} \leq Q^{\pi}$$

Conservative Q-Learning

$$L_{CQL} = \max_{\mu} [\alpha (\mathbb{E}_{s \sim D, a \sim \mu(\cdot|s)} Q_{\phi}(s, a)) - \alpha \mathbb{E}_{(s,a) \sim D} Q_{\phi}(s, a) + \\ + \frac{1}{2} \mathbb{E}_{(s,a,r,s') \sim D} [r + \gamma \mathbb{E}_{a' \sim \pi} Q_{\phi}(s', a') - Q_{\phi}(s, a)]^2] \rightarrow \min_{\phi}$$

It can be shown that for large enough α

$$\mathbb{E}_{\pi} Q_{\phi} \leq \mathbb{E}_{\pi} Q^{\pi}$$

Conservative Q-Learning

$$L_{CQL} = \max_{\mu} [\alpha (\mathbb{E}_{s \sim D, a \sim \mu(\cdot|s)} Q_{\phi}(s, a)) + R(\mu)] - \alpha \mathbb{E}_{(s,a) \sim D} Q_{\phi}(s, a) + \\ + \frac{1}{2} \mathbb{E}_{(s,a,r,s') \sim D} [r + \gamma \mathbb{E}_{a' \sim \pi} Q_{\phi}(s', a') - Q_{\phi}(s, a)]^2 \rightarrow \min_{\phi}$$

Conservative Q-Learning

Common choice $R(\mu) = \mathbb{E}_{s \sim D} H(\mu(\cdot | s))$

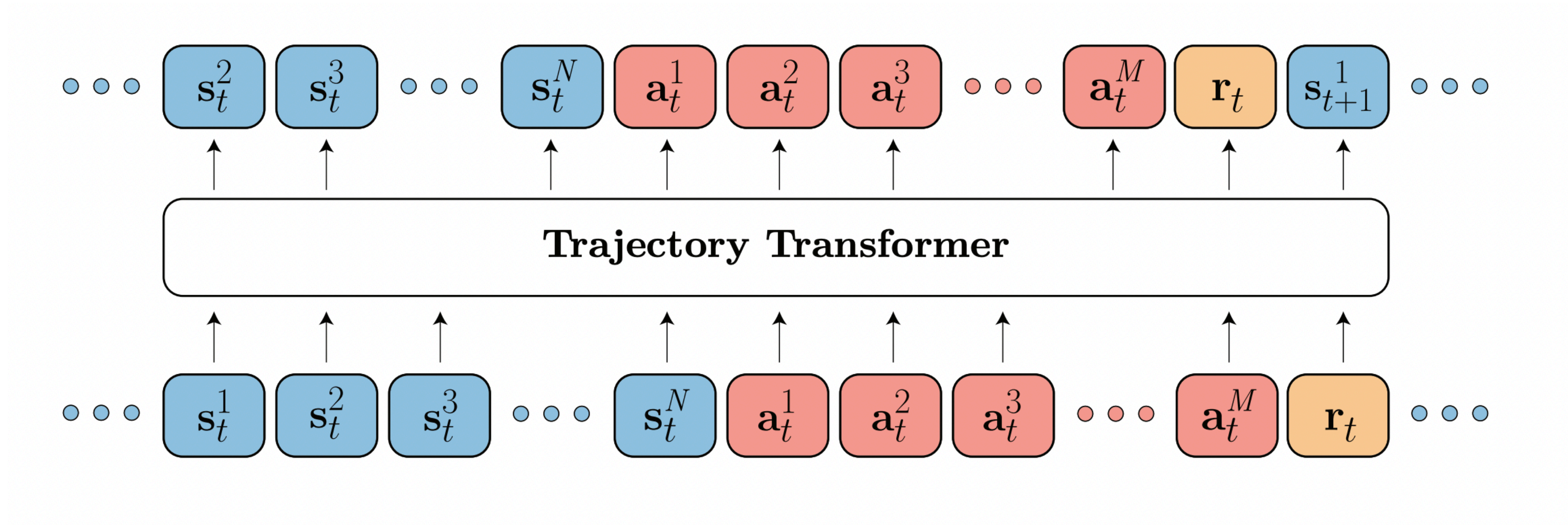
$$\mathbb{E}_{s \sim D, a \sim \mu(\cdot | s)} [Q_\phi(s, a) + H(\mu(\cdot | s))] \rightarrow \max_{\mu} \implies \mu(a | s) \propto \exp(Q_\phi(s, a))$$

$$L_{CQL} = \alpha(\mathbb{E}_{s \sim D} \log \sum_a \exp(Q(s, a))) - \alpha \mathbb{E}_{(s, a) \sim D} Q_\phi(s, a) \\ + \frac{1}{2} \mathbb{E}_{(s, a, r, s') \sim D} [r + \gamma \mathbb{E}_{a' \sim \pi} Q_\phi(s', a') - Q_\phi(s, a)]^2 \rightarrow \min_{\phi}$$

1. Learn Q_ϕ optimising L_{CQL} on offline data
2. Update policy with SAC-style entropy regularization:

$$\mathbb{E}_{s \sim D, a \sim \pi_\theta} [Q_\phi(s, a) - \log \pi_\theta(a | s)] \rightarrow \max_{\theta}$$

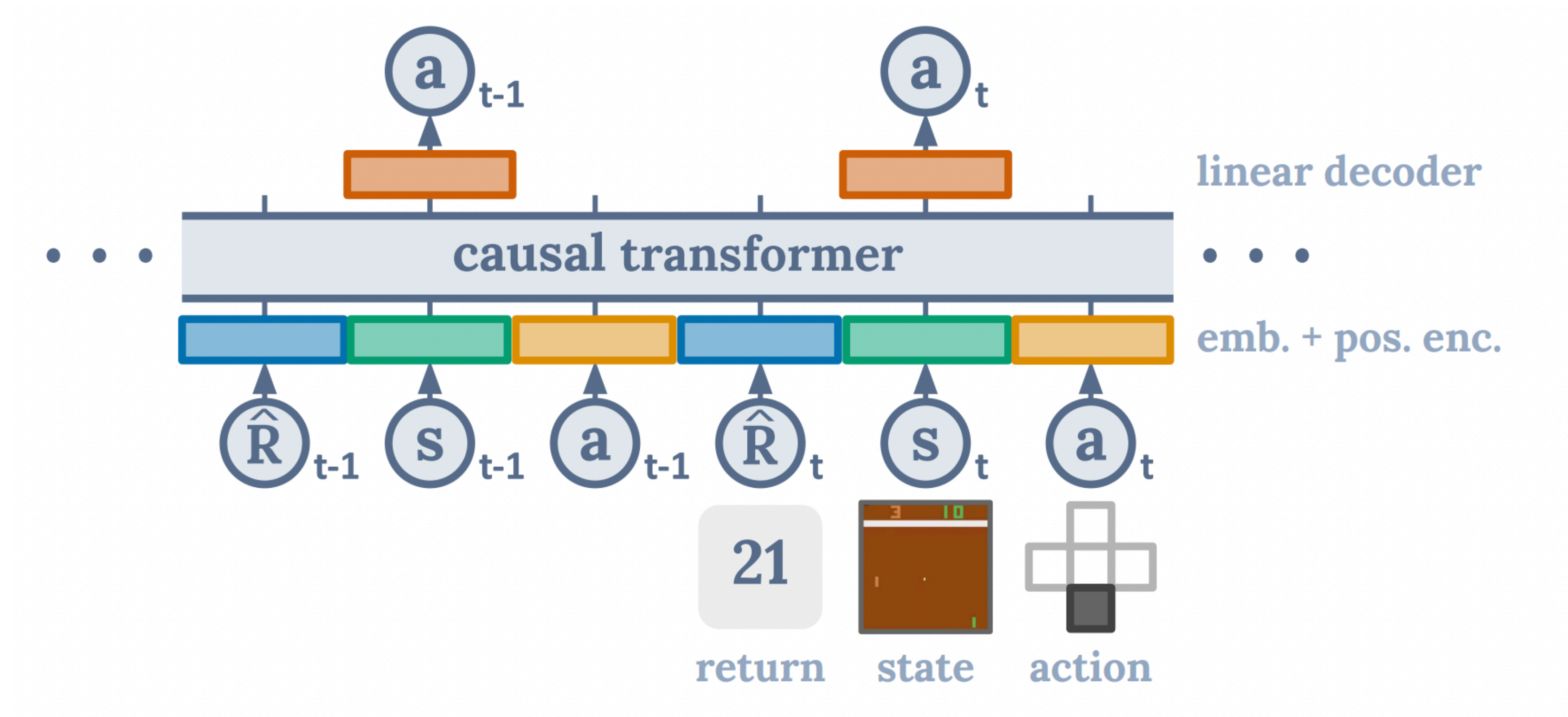
Trajectory Transformer



$$\mathcal{L}(\tau) = \sum_{t=1}^T \left(\sum_{i=1}^N \log P_{\theta}(s_t^i | \mathbf{s}_t^{<i}, \boldsymbol{\tau}_{<t}) + \sum_{j=1}^M \log P_{\theta}(a_t^j | \mathbf{a}_t^{<j}, \mathbf{s}_t, \boldsymbol{\tau}_{<t}) + \log P_{\theta}(r_t | \mathbf{a}_t, \mathbf{s}_t, \boldsymbol{\tau}_{<t}) \right),$$

Planning with Beam Search using sum of rewards

Decision Transformer



Recap: TD3

1. Deterministic Policy Gradient:

$$\mathbb{E}_{s \sim D}[Q_{\phi}(s, \pi_{\theta}(s))] \rightarrow \max_{\theta}$$

2. Clipped Double-Q Learning:

$$y = r + \gamma \min_{i=1,2} Q_{\phi_i^-}(s', \mu_{\theta^-}(s'))$$

3. Delayed Policy Updates: Update the policy (and target networks) less frequently than the Q-function.

$$y = r + \gamma \min_{i=1,2} Q_{\phi_i^-}(s', \mu_{\theta^-}(s'))$$

4. Target Policy Smoothing:

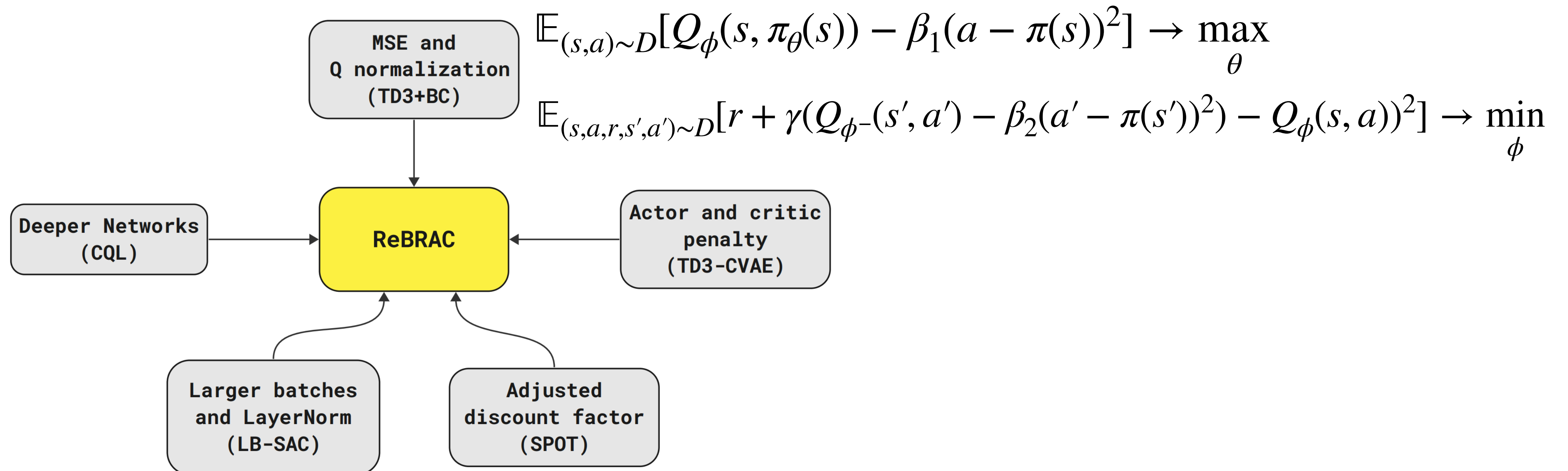
$$y = r + \gamma \min_{i=1,2} Q_{\phi_i^-}(s', a'), a' = \mu_{\theta^-}(s) + \varepsilon',$$
$$\varepsilon' \sim \text{clip}(\mathcal{N}(0, \sigma' I), -c, c)$$

The Minimalist Approach to Offline RL

TD3+BC:

$$\mathbb{E}_{(s,a) \sim D} [\lambda Q_{\phi}(s, \pi_{\theta}(s)) - (a - \pi_{\theta}(s))^2] \rightarrow \max_{\theta}$$

ReBRAC:



Not Covered Yet

1. Uncertainty-Based Offline Reinforcement Learning with Diversified Q-Ensemble
2. Anti-Exploration by Random Network Distillation

Background

1. Offline Reinforcement Learning: Tutorial, Review, and Perspectives on Open Problems
2. RL course by Sergey Levine, Lectures 15-16, Playlist
3. CORL: Research-oriented Deep Offline Reinforcement Learning Library

Thank you for your attention!